

Leukemia Cell Classification Using Deep Learning Model

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Introduction

- Leukemia is a serious blood cancer that requires early and accurate diagnosis.
- Manual microscopic diagnosis is time-consuming and highly dependent on expert pathologists.
- This project develops an AI-based automated system using deep learning (ResNet) to classify leukemia from blood smear images.

Objective

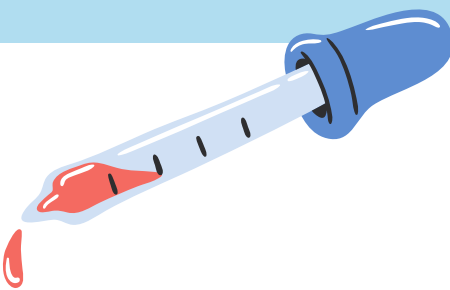
- Develop a real-time automated deep learning system to classify leukemia subtypes (AML, ALL, CLL, CML).
- Improve model performance using preprocessing and data augmentation techniques.
- Ensure reliable clinical performance using standard evaluation metrics (accuracy, precision, recall, F1-score).

Gap Analysis

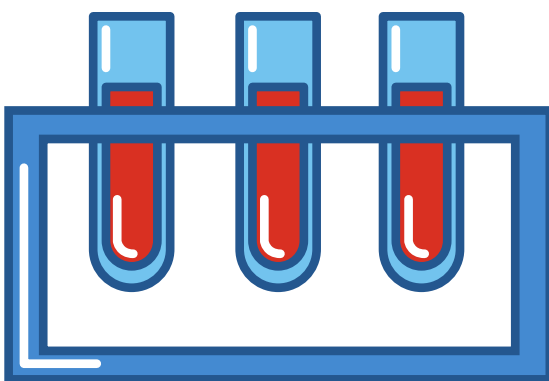
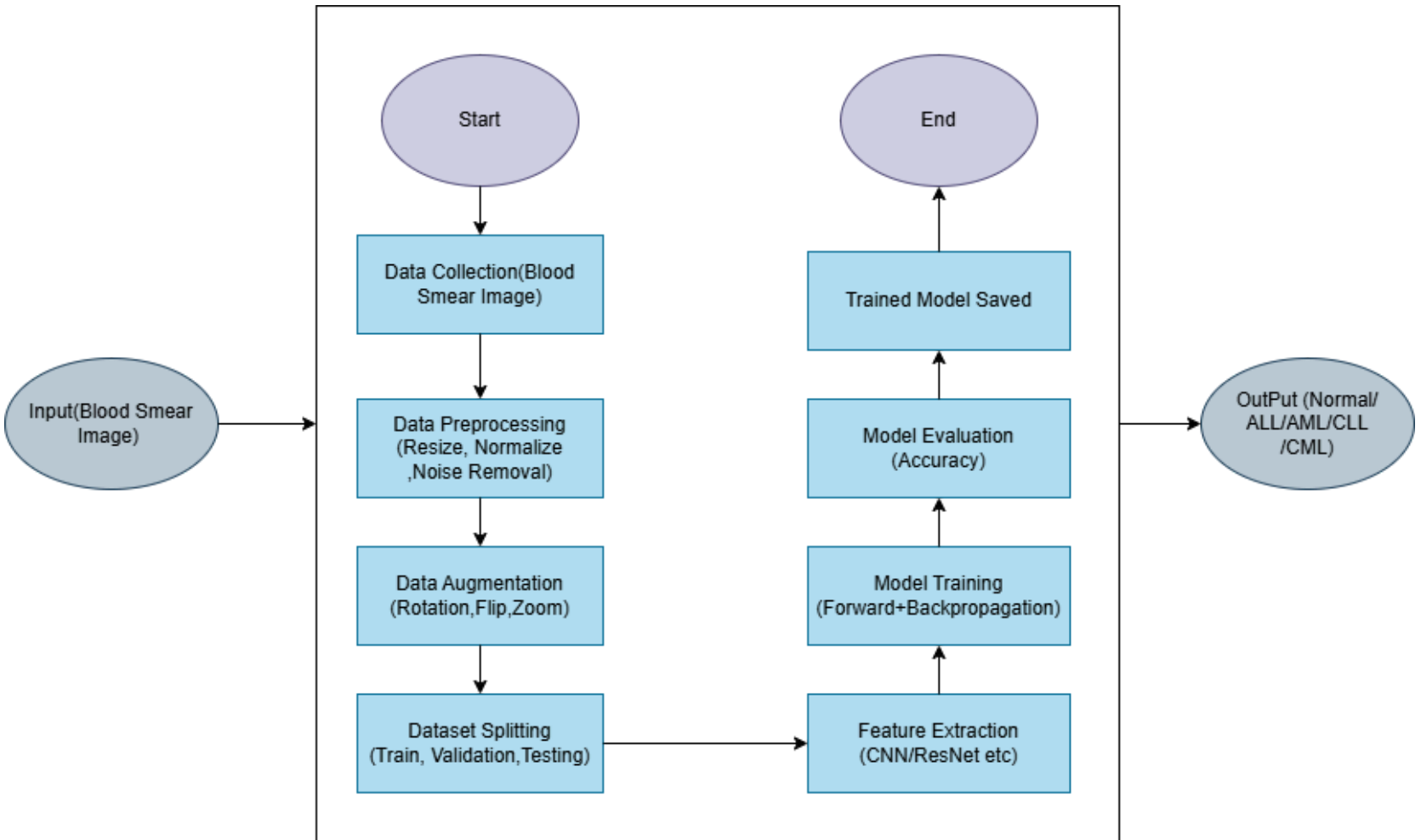
Paper Name	Key Findings	Tools & Techniques	Identified Gap
Medical image-based detection of Acute Lymphoblastic Leukemia	MobileNetV2: 99.69% accuracy	Dataset: ALL image dataset Models: MobileNetV2, Custom CNN.	Limited Dataset, Focused only on ALL, Model tested on static dataset
Computer-Based Leukemia Diagnosis System	K-NN give the best accuracy 86%	Dataset: Total 260 images Model: K-NN, DT, SVG Image Segmentation	Small dataset, Low diversity
Leukemia Subtype detection using CNN from microscopic image	88.25% for binary classification and 82% for classifying all leukemia subtypes	Dataset: ALL-IDB and ASH Image Bank.	optimize model using advanced deep learning models for higher multi class accuracy.
Unintentional detection of leukemias with complete blood count	Incidental detection in asymptomatic leukemias, Risk of overdiagnosis and overtreatment	Use CBC, supplemented ,Use clinical decision aids for shared decision-making	No formal recommendations for systematic screening in hematology
Classification of leukemia cell by CNN	CNN achieved 96.6% accuracy for binary classification	Dataset: ALL-IDB1 Method: Data Augmentation and implementation by MATLAB.	Focused only on ALL, limited clinical and real-world validation.Focused , limited clinical and real-world validation.
Hybrid Approach for ALL Classification	ResNet50, SVM, and RF perform best for the kind of work	Deep Models: ResNet152, VGG16, VGG16, DenseNet121, MobileNetV2	Manual Detection slow, insufficient data for DL model, manually time consuming

Key Innovation

This project develops an automated AI-based system using ResNet to classify leukemia subtypes from microscopic blood smear images. It integrates advanced feature extraction with data augmentation to improve accuracy and robustness. The system provides a reliable decision-support tool for medical diagnosis.

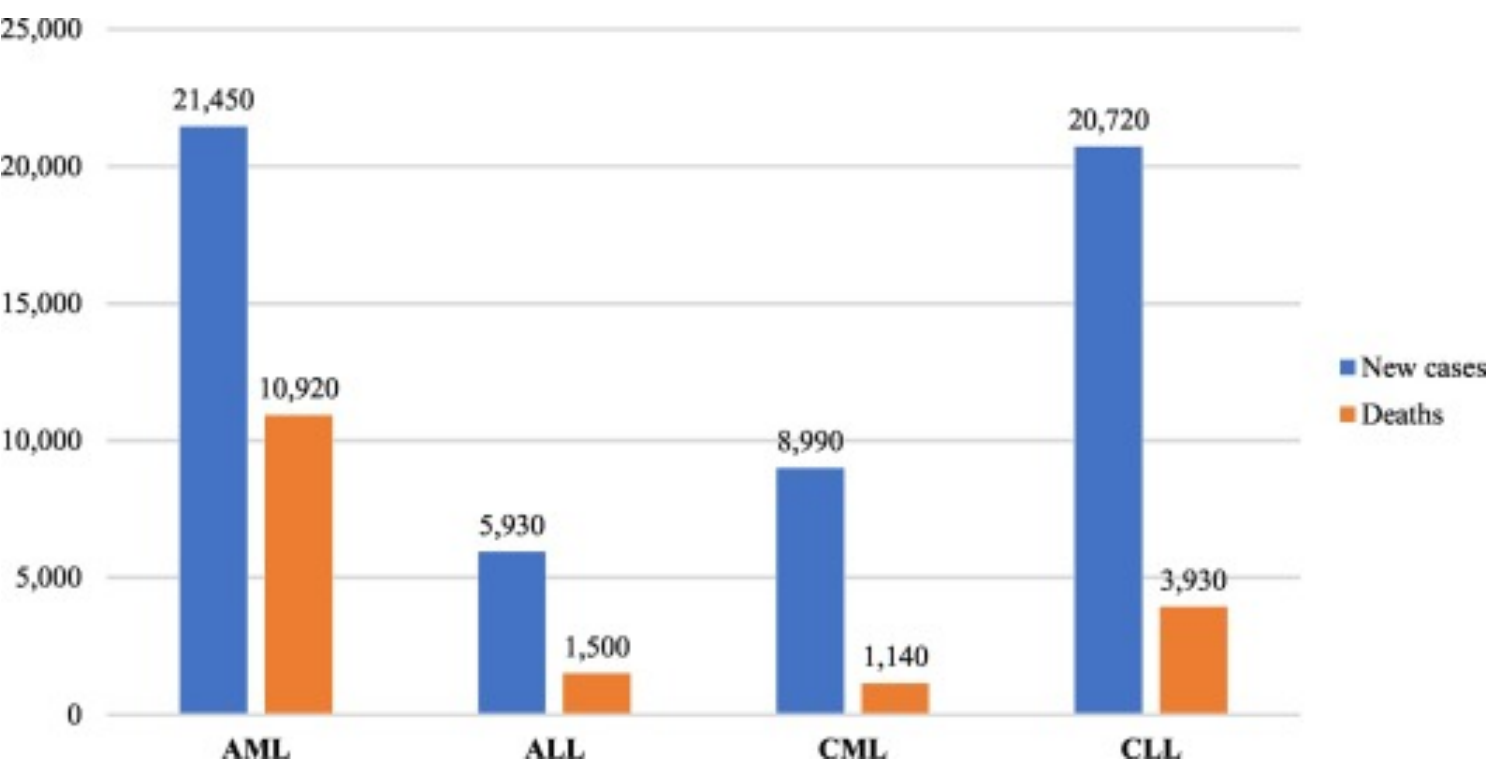


Methodology



Applications

The system enables automatic classification of AML, ALL, CLL, and CML, assisting hematologists in faster diagnosis. It reduces manual workload and diagnostic errors. It also has potential for integration into hospital diagnostic support systems.



Cancer Facts & Figures of 2021